

ZHENGMIN PIAO

Hefei, China

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RESEARCH INTERESTS

Machine Learning, Stochastic Modeling, Optimization and Decision-Making, with applications to Biomedical Data and Data-Driven Systems.

EDUCATION

University of Science and Technology of China (USTC), Hefei, China Aug 2023 – Present
B.S. in Statistics, Minor in Finance (Expected Graduation: Jun 2027)

RESEARCH EXPERIENCE

Research Assistant, Michigan State University Jul 2026 – Present
Supervisor: Prof. Yuying Xie, Department of Computational Mathematics, Science and Engineering

- Contributing to a research project on scaling behavior in plant single-cell transcriptomics, with a current focus on reproducing and benchmarking computational methods for cell-type annotation and marker identification.
- Built and evaluated single-cell RNA-seq analysis workflows in R and Python, including data preprocessing, normalization, dimensionality reduction, clustering, marker-gene analysis, and reference-based cell-type annotation using Seurat and Scanpy.
- Reproduced and benchmarked annotation approaches including HCMarker, PCmaster, and ScType across plant scRNA-seq datasets, and performed statistical analyses to assess marker specificity, annotation consistency, and robustness across datasets.

Undergraduate Thesis Research, USTC May 2026 – Present
Advisor: Prof. Ning Zhu, Department of Management Science
Distribution Estimation and Routing Optimization for Stochastic Service Times with Contextual Information

- Investigating conditional distribution estimation for stochastic service times using contextual information, with emphasis on finite-sample performance, nonlinear dependence, and model misspecification.
- Studying parametric, semiparametric location-scale, and nonparametric approaches for estimating conditional service-time distributions and incorporating the resulting uncertainty into stochastic and robust routing models.
- Designing out-of-sample computational experiments to evaluate how uncertainty estimation affects downstream routing decisions in terms of expected reward, feasibility, and tail risk.

Research Assistant, Anhui Provincial Major Social Science Project Sep 2025 – Dec 2025
Supervisor: Prof. Jiaquan Yao, Department of Statistics and Finance

- Analyzed firm-level panel data using econometric and multivariate statistical methods to study the relationship between patient capital, R&D investment, and innovation outcomes.

- Applied fixed/random-effects panel regression, PCA, and ridge regression to address high-dimensional and small-sample settings and quantify various effects on innovation performance.

Intern, Organization Department, Yanbian Prefecture Government Jul 2025 – Aug 2025
Yanbian, Jilin, China

- Processed multi-year incident datasets (fires, traffic accidents, structural collapses, and floods) using Python and developed optimization models for surveillance camera placement.
- Formulated and solved constrained optimization models using Gurobi/CPLEX; implemented Lagrangian relaxation and ADMM for large-scale instances, with robustness considerations for camera reliability.
- Conducted simulation-based evaluation of coverage, cost efficiency, and robustness; the resulting deployment plan was reported to reduce system cost by approximately 40%.

HONORS & COMPETITIONS

Meritorious Winner, Mathematical Contest in Modeling (MCM/ICM), Problem C	2026
First Prize, Anhui Division, China Undergraduate Mathematical Contest in Modeling (CUMCM)	2025
Third Prize, Anhui Division, National Undergraduate Mathematics Competition	2024

VISITING PROGRAMS

Global Leadership Visit and Exchange Program, Singapore Management University 2025

Project: AI-driven quantitative investment strategy design using convolutional neural networks and NLP-based sentiment analysis on financial data.

LEADERSHIP & TEACHING EXPERIENCE

President, Student Union, School of Management	Apr 2025 – Apr 2026
Student Representative, AI Computing Platform (USTC)	Apr 2026 – Jul 2026
Teaching Assistant, Probability Theory and Mathematical Statistics	Aug 2026 – Present

SKILLS

Programming: Python, R, SQL, Julia, C++

Optimization Tools: Gurobi, CPLEX

Languages: Chinese (native), Korean (native-level), English (CEFR C1)